VASANTHAGANESH R

Backend Java Engineer · Enterprise Microservices & Scalable Systems

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PROFESSIONAL SUMMARY

Backend Java Engineer with 2+ years of enterprise experience at **Virtusa** engineering high-throughput Spring Boot microservices and transactional banking workflows for **Citi's global banking platform**. Deep expertise in Core Java (8/17), ACID compliance, relational query tuning, and contract-first REST APIs where zero data loss and 99.99% availability are non-negotiable. **Oracle Certified Associate (OCA) Java Programmer** and **Claude AI Certified**, combining battle-tested BFSI engineering rigor with modern AI-augmented developer velocity and full-stack product execution.

TECHNICAL SKILLS

Backend & Core Java:	Core Java (8/17), Spring Boot, Spring Data JPA, Hibernate, Microservices, REST APIs, Concurrency, Maven, JUnit 5, Mockito
Databases & Integrity:	PostgreSQL, MySQL, Oracle DB, Schema Design, B-Tree Indexing, Query Tuning, Strict ACID Compliance
AI & Emerging Tools:	Claude API, Anthropic Prompt Engineering, Context Memory Architectures, Python, LLM Workflows
Frontend & Mobile:	React, React Native, TypeScript, Node.js, Flutter, Cross-Platform Mobile Architecture
DevOps & 3D Tools:	Git, IntelliJ IDEA, Docker, Postman, Linux, Blender, Unity, Spatial Systems

PROFESSIONAL EXPERIENCE

Associate Engineer — Backend Java | Virtusa (Client: Citi) — Aug 2023 – Sep 2025 | Puducherry, India

- **Architected high-throughput Spring Boot microservices** and contract-first REST endpoints powering Citi's retail banking platform, sustaining **15,000+ daily transactions with 99.99% uptime** under strict BFSI compliance standards.
- **Enforced strict ACID transactional boundaries**, deterministic rollback handlers, and idempotent endpoints across financial transactions, guaranteeing **0% financial data loss** during network partitions and peak clearing windows.
- **Tuned relational database execution plans**, indexing strategies, and Hibernate/JPA entity mappings on **Oracle DB**, slashing query response times by **38% during peak settlement hours**.
- **Systematized unit and integration test pipelines** using **JUnit 5 and Mockito**, driving test coverage past **85%** and decreasing regression defects by **25%** prior to staging handoff.
- Collaborated in distributed Agile sprints with global cross-functional squads to deliver mission-critical banking features on schedule.

Independent Backend & Systems Engineer | Self-Employed — Oct 2025 – Present | Remote

- **Precision Tools QC & Report Manager:** Spearheaded an industrial mobile QC suite in React Native with automated PDF generation, cutting shop-floor inspection times by **90% (from 5-minute paper checklists down to a 20-second digital workflow)** across 500+ daily manufactured parts.
- **Precision Tools B2B Corporate Platform:** Delivered a high-performance React B2B showcase featuring aggressive asset optimization, attaining a **98+ Google Lighthouse performance score and sub-800ms page load times**.
- **Daily Vegetable Price Tracker:** Deployed a community Android utility delivering verified real-time market rates for **1,200+ local families and street vendors** across Puducherry, curbing middleman price markups by **15–20%**.
- Managed complete engineering lifecycles independently: client discovery, relational schema design, API architecture, implementation, and cloud deployment.

Technical Internships | VEI Technology & Askan Technology — 2022 | Puducherry, India

- **VEI Technology (IoT Systems):** Built a hardware-to-cloud sensor data pipeline streaming telemetry at **1-second intervals** to a cloud monitoring dashboard.
- **Askan Technology:** Completed in-plant engineering training in cross-platform mobile application development with Flutter, delivering **2 functional client UI modules**.

FEATURED SYSTEMS & AI PROJECTS

AI Reflection Journal React · TypeScript · AI Prompts · Mental Wellness · Privacy github.com/vasanthaganesh/AI-Reflection-Journal ↗

- Designed an introspective digital sanctuary for individuals navigating emotional overwhelm and anxiety who struggle to express thoughts out loud.
- Formulated structured cognitive restructuring prompt chains with **sub-second inference latency** and client-side encryption for **100% user data privacy**.

AI Companion Python · LLM Workflows · Persistent Context Memory · TypeScript github.com/vasanthaganesh/AI_Companion ↗

- Constructed an empathetic conversational partner equipped with **persistent contextual memory** retaining **30+ dialogue turns** per session to alleviate social isolation.
- Maintained conversation nuances and personal context over time, delivering an attentive, safe sounding board and empathetic presence.

Citi Enterprise Banking Platform Java 8/17 · Spring Boot · Microservices · Oracle DB · REST APIs Virtusa Client Project

- Core transactional banking pipelines processing high-volume institutional ledger operations under strict regulatory audits and **zero downtime tolerances**.

ACADEMIC & ENGINEERING PROJECTS

3D Interactive Virtual Campus Visualization Blender · Unity · 3D Modeling · Interactive Systems · C# Final Year Capstone Project

- Constructed a full-scale **3D interactive virtual campus exploration platform** showcasing the complete architectural infrastructure of the college campus.
- Refactored complex 3D building meshes by **65% (from 1.2M polygons down to 420K)** while integrating camera collision physics and spatial wayfinding to sustain a smooth **60 FPS walkthrough**.

Appverse Application Android · Java · Firebase Realtime DB Campus Academic Portal

- Implemented a campus collaboration mobile app serving **800+ students and faculty** with real-time assignment submissions, announcements, and push notifications.

AudioBook Application Android · Java · Accessibility APIs Assistive Mobile App

- Created an assistive audiobook reading tool tailored for learning-disabled students with offline audio caching and variable-speed playback across **100+ audio chapters**.

Abstraction-Based Text Summarization Python · NLP · Flask · Web Application NLP Analytical Tool

- Benchmarked NLP abstractive text summarization models, condensing **50+ page legal and investigative reports** into concise briefs and reducing officer review time by **60%**.

CERTIFICATIONS

- **Oracle Certified Associate (OCA)**, Java SE 8 Programmer (Core Java & Concurrency) — Flagship Credential
- **Claude AI Certified** — Anthropic Prompt Engineering & Generative Workflows
- **Talent-Next Java Full Stack Certification** — Wipro / NASSCOM
- **JavaScript Essentials 1** — Online Certification
- **Linux Training** — Spoken Tutorial, IIT Bombay

HONORS & HACKATHONS

- 🥉 **3rd Prize** — AI-HACK Project Development Competition (MVIT)
- 🥉 **3rd Prize** — SCIMIT-23 National Level Project Expo
- 🥇 **1st Prize** — Paper Presentation, Christ College of Engineering
- 🥇 **1st Prize** — Blind Coding, Pondicherry Technological University
- 🏅 **Participant** — Smart India Hackathon (SIH 2022) & Virtusa Jatayu

EDUCATION & VITALS

Bachelor of Technology (B.Tech) — Information Technology | Manakula Vinayagar Institute of Technology Puducherry, India | Graduated 2023 (CGPA: 8.32 / 10.0)

Languages: English, Tamil • **Availability:** Immediate • **Relocation:** Open to relocation globally